

Calibration Split Instability in Mixed-Precision LLM Quantization

Anonymous Submission

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Abstract

Mixed-precision post-training quantization (PTQ) often turns a small calibration set into a high-stakes allocation decision: which modules should receive scarce high-bit precision? We argue that a key failure mode is not only low final accuracy, but behavior associated with *decision instability*: equally plausible calibration splits can induce materially different sensitivity rankings and therefore different allocations under the same bit budget. Calibration Split Instability (CSI) provides a signal for estimating calibration-induced instability in PTQ systems, rather than a standalone metric or a new quantization backend. CSI audits seed-pair agreement using rank correlation, top- k overlap, positive-set overlap, bootstrap trend tests, and permutation-null tests before a calibration-derived allocation is promoted from diagnostic evidence to a stronger method claim. Across Qwen2.5 same-prompt-pool artifacts, increasing calibration size correlates with higher stability: for Qwen2.5-1.5B, mean Spearman rises from 0.1410 at $n = 2$ to 0.4918 at $n = 8$, with positive bootstrap gain intervals and Holm-adjusted permutation $p \leq 0.00015$. A stricter second-pool SmolLM2-360M replication shows the same trend from $n = 4$ to $n = 16$, with mean Spearman improving from 0.3133 to 0.6959 and Holm-adjusted permutation $p = 0.0005999$. We further separate quantization backends from decision-layer evidence by confronting native PTQ baselines on full Qwen2.5-1.5B downstream benchmarks: on MMLU (14,042 examples), FP16/AutoAWQ/GPTQModel reach 0.5864/0.5648/0.5362, and on GSM8K (1,319 examples), 0.0811/0.0788/0.0728. Local GPU fake-quant smokes exercise the CSI/consensus allocation path without upgrading it to a SOTA claim. The result is an audit framework that characterizes calibration regime shift, provides indication of allocation risk, and routes each claim to the evidence required before stronger downstream claims can be asserted.

Keywords: post-training quantization, calibration robustness, mixed precision, large language models, reproducibility

1 Introduction

Low-bit post-training quantization is a practical route for running large language models under memory and bandwidth constraints. In mixed-precision PTQ, however, compression quality depends on a discrete decision made before the final task score is known: which modules deserve scarce high-bit budget? Uniform quantization avoids that decision by treating every module equally, but calibration-driven allocation must infer a module ordering from a small prompt sample. When the ordering changes under another plausible split, PTQ failure can be viewed as being associated with decision instability under calibration shift and the same resource constraint.

This makes calibration instability a reliability concern for any method whose claim depends on *which* layers, blocks, or channels are protected. Existing PTQ backends can be strong while still not explicitly modeling instability boundaries in calibration-driven allocation: they may optimize reconstruction, rescale activations, learn transforms, or protect salient channels, but they do not necessarily characterize whether the upstream calibration-derived allocation would be similar under a nearby calibration draw. Our novelty anchor is therefore the following: *we treat calibration-driven PTQ allocation as an auditable process whose claims are checked against stability, retention, and resource evidence*. A low-bit allocation is more credible when the ranking is stable across plausible splits, the

retained model behavior passes an evaluation threshold, and the resource budget is respected.

Many calibration-driven pipelines answer this question by estimating module sensitivity on a small prompt set, ranking modules, and protecting those that appear most sensitive. This decomposition is convenient, but it hides a failure mode. If two small calibration samples drawn from the same prompt pool produce different sensitivity rankings, a single-split allocation can look principled while encoding sampling noise. Such instability matters even when the final quantization backend is strong: GPTQ, AWQ, SmoothQuant, OmniQuant, QuaRot, and SpinQuant define powerful ways to transform or quantize weights and activations [1, 3, 7, 8, 12, 14], but a mixed-precision allocation layer still benefits from evidence that its calibration-derived ranking is consistent enough to support stronger claims.

This paper studies *calibration split instability* (CSI) as such an instability signal. CSI is not a new quantization backend and not a direct claim of state-of-the-art PTQ quality. It is a scoring function used to quantify calibration-induced instability before calibration-derived decisions are used to allocate precision. CSI is deliberately positioned between calibration-data studies and final task-retention benchmarking: calibration data can affect compression outcomes [13], and benchmark toolkits emphasize standardized evaluation [4]; CSI targets the

CSI Decision Feedback Loop
Calibration stability as a policy-driven allocation audit

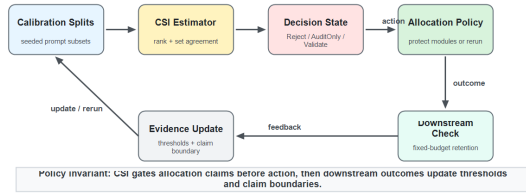


Figure 1: CSI audit loop. Calibration splits feed an instability score that informs an allocation policy before stronger claims are made. Downstream outcomes do not retroactively prove calibration stability; they update the claim boundary and future thresholds.

intermediate ranking/allocation instability that can make those outcomes difficult to interpret.

Contributions. We make three claims. First, we characterize calibration regime shift as an instability pattern associated with mixed-precision PTQ allocation: at small calibration sizes, a single calibration ranking can encode sampling noise while still appearing methodologically precise. Second, we propose CSI as a scoring function that provides indication of this instability and supports operational states such as reject, audit-only, or downstream-validation candidate. Third, we organize the current evidence into a failure-space audit: native PTQ backends, CSI allocation smokes, low-bit scaffolds, and failure heatmaps are separated by what they can and cannot support, reducing the risk that metric evidence is misreported as downstream quality evidence.

2 Problem Setup

Let a transformer have quantizable modules $\mathcal{M} = \{m_1, \dots, m_L\}$. A mixed-precision allocation is $a = (b_1, \dots, b_L)$, where each bit width b_i belongs to a finite set $\mathcal{B}$, such as $\{4, 8\}$. Let $C = \{C_1, \dots, C_S\}$ be calibration splits sampled from a prompt pool. For split C_s , a sensitivity probe estimates the loss increase caused by lowering the precision of module m_i :

$$\hat{\Delta}_{i,s} = \ell(f_{a^{(i\downarrow)}}; C_s) - \ell(f_{a^{ref}}; C_s), \quad (1)$$

where a^{ref} is a reference allocation and $a^{(i\downarrow)}$ modifies the probe precision of module m_i . A single-split method ranks modules using one column of $\hat{\Delta}$. CSI asks whether those columns agree.

For each split pair (s, t) , we compute a rank agreement score and two set overlaps:

$$\rho_{s,t} = \text{Spearman}(\hat{\Delta}_{:,s}, \hat{\Delta}_{:,t}), \quad (2)$$

$$J_{s,t}^k = \frac{|\text{TopK}_s(k) \cap \text{TopK}_t(k)|}{|\text{TopK}_s(k) \cup \text{TopK}_t(k)|}, \quad (3)$$

$$J_{s,t}^{pos} = \frac{|P_s \cap P_t|}{|P_s \cup P_t|}, \quad (4)$$

where $P_s = \{i : \hat{\Delta}_{i,s} > 0\}$. We report pairwise means and bootstrap confidence intervals over seed pairs. The CSI

gate also compares the smallest and largest calibration sizes by independent bootstrap gain intervals and a Monte-Carlo label-shuffle null with Holm correction.

Intended use. The statistic above is most relevant for claims that use calibration to choose a non-uniform allocation. If an allocation ignores calibration, CSI may be less informative; if a method uses calibration only inside a backend reconstruction objective, CSI audits only the external allocation or protection decision. This restriction makes CSI a calibration-instability signal for allocation studies, not a universal quality metric for every PTQ algorithm.

Composite audit signal. Within this decision class, CSI is a composite scoring function rather than a theoretically exact boundary estimator. A calibration-driven allocation policy often depends on three observable decisions: the relative module order, the protected top- k set, and the sign of measured sensitivity. Reporting (ρ, J^k, J^{pos}) characterizes these complementary views: rank agreement, protected-set overlap, and sign consistency. The tuple therefore indicates allocation-changing perturbations without claiming complete detection or final task conclusions.

Baseline class framing. The comparison classes are not interchangeable. GPTQ, AWQ, SmoothQuant, OmniQuant, QuaRot, and SpinQuant are backend families; uniform quantization is a no-allocation baseline; and budget-matched random allocation removes the calibration term. CSI addresses the remaining class: pipelines that use calibration to decide *which* modules to protect. It is therefore reported as an instability signal alongside native PTQ backends, not as their replacement.

3 CSI as an Instability Score

CSI summarizes calibration evidence as an instability score that can inform operational decisions. We use three non-exclusive reporting states: REJECT for evidence with high measured instability, AUDITONLY for evidence that supports a bounded audit claim but lacks direct downstream retention, and VALIDATE for candidates that have also passed fixed-budget downstream evaluation. These states are implementation choices for claim management, not theoretically exact stability regions.

The stability part of the operator can be used as a hard gate or as a soft allocation risk. Let

$$\mathcal{G}_{stab}(a; C) = w_\rho \bar{\rho} + w_k \bar{J}^k + w_p \bar{J}^{pos}. \quad (5)$$

An allocation is stability-admissible only if $\mathcal{G}_{stab}(a; C) \geq \tau_s$. A smooth risk version is

$$\phi_s(a) = \log(1 + \exp(\gamma_s(\tau_s - \mathcal{G}_{stab}(a; C)))) \quad (6)$$

Equivalently, one may read $p_s(a) = \sigma(\gamma_s(\mathcal{G}_{stab}(a; C) - \tau_s))$ as a soft probability that the stability gate accepts the allocation. This probabilistic view is useful operationally: a saturated low p_s indicates calibration collapse, while high p_s with poor retention indicates that stable rankings are not enough

for quality. The full constrained calibration objective can be written as

$$\begin{aligned}
\min_{a \in \mathcal{B}^L} \quad & \mathcal{J}(a) = \mathcal{L}_{task}(a; D) + \lambda_c \mathcal{L}_{cal}(a; C) \\
& + \lambda_s \phi_s(a) + \lambda_u \mathcal{R}_{uncert}(a; C) \\
& + \lambda_b \mathcal{R}_{budget}(a) \\
\text{s.t.} \quad & \sum_i \text{mem}(m_i, b_i) \leq M, \quad \mathcal{G}_{ret}(a; D_{val}) \geq \tau_r.
\end{aligned} \tag{7}$$

$$\tag{8}$$

This objective separates evidence into the same practical reporting states. If the stability score is weak, the calibration evidence is treated as under-supported. If the stability score is stronger but direct retention is missing, the evidence remains audit-only. If both stability and retention pass under a fixed budget, the allocation can be discussed as a downstream-validated candidate. The current paper establishes the first two parts of this pipeline at scale and reports matched native-PTQ downstream retention rows. It does not yet claim that a CSI-informed allocation changes those downstream rows end to end.

Propagation argument. CSI matters because calibration instability can propagate through the allocation map. Let π_s be the module order induced by split C_s and let $A(\pi_s, M)$ be the set of modules protected under budget M . Low split agreement can make $A(\pi_s, M)$ and $A(\pi_t, M)$ differ under the same budget, changing which modules are protected or degraded. Thus calibration size $\downarrow$ can be associated with rank instability $\uparrow$, allocation variance $\uparrow$, and downstream-risk exposure $\uparrow$. High CSI does not establish task success; low CSI suggests rerunning calibration, enlarging the prompt pool, falling back to a uniform/backend-native policy, or reporting only an audit failure.

Monotonic stabilization. Because $\mathcal{B}^L$ is finite, a search that accepts only improving moves exhibits monotonic behavior under bounded updates and settles into a local stable region for the chosen neighborhood. This is a stabilization observation, not a SOTA-quality theorem.

4 Experimental Protocol

Prompt pools and models. The Qwen2.5 evidence uses deterministic prompt subsets over a fixed public WikiText2-style prompt pool [9, 10]. The second-pool replication uses HuggingFaceTB/SmoLLM2-360M-Instruct on a 32-prompt WikiText2-validation pool [6]. Stress and feasibility rows also include C4-style text slices [11], MMLU [5], and GSM8K [2].

Metrics. For each calibration size n , six deterministic prompt seeds are evaluated. This yields 15 seed pairs per n . We report mean Spearman, top-20 Jaccard, and positive-set Jaccard with bootstrap confidence intervals. Trend gates compare the smallest and largest n values by bootstrap mean-gain intervals and bootstrap win probabilities. Null gates shuffle the small/large n labels and apply Holm correction across the three audited metrics.

Native PTQ downstream retention. To avoid a purely diagnostic evaluation, we include matched downstream rows for Qwen2.5-1.5B FP16, AutoAWQ, and GPTQModel variants. MMLU uses all 14,042 test examples across 57 subjects, and GSM8K uses all 1,319 examples. For each PTQ candidate, we report accuracy, Wilson intervals, paired bootstrap candidate-minus-FP16 deltas, and discordant-pair counts from the same task fixture. These rows confront hard native PTQ baselines, but they are not CSI-informed allocation rows.

Claim boundary. The evidence ledger treats a result as paper-facing only when it has a JSON artifact, a generated Markdown gate report, a runnable command in the runbook, and a claim-matrix boundary. Results outside that boundary are discussed only as future work or local feasibility evidence.

Detect-classify-route protocol. Following the claim boundary, experiments are organized around the instability score rather than around a flat leaderboard. The first layer *detects* calibration regime shift by measuring how rank and set agreement change with calibration size. The second layer *classifies* evidence into practical reporting states such as REJECT, AUDIT-ONLY, or VALIDATE-candidate using the CSI score, budget checks, and failure hooks. The third layer *routes* claims to the evidence they require: native full-benchmark PTQ rows test downstream retention for backend baselines, local allocation smokes test the allocation path, and extreme-bit scaffolds mark regimes that still require direct same-budget downstream evaluation.

Decision-system audit. To make the empirical story falsifiable, a claim is promoted only if the corresponding artifact measures the required regime under the stated boundary. This prevents a local PPL smoke from being treated as a full MMLU/GSM8K win, while still making visible where CSI changes allocation behavior and where the submission-critical rows remain. In this sense, CSI predicts *claim risk* rather than downstream accuracy directly: low split agreement predicts that an allocation claim is unsafe to promote, while high agreement only permits the next validation step.

Policy feedback loop. In Figure 1, a calibration pool produces a CSI score, the score suggests a reporting state, the allocation policy is executed only for better supported states, and downstream outcomes update thresholds, claim maps, or rerun decisions.

Mechanism chain. The intended explanatory chain is not “more components give better numbers.” It is *calibration sampling* $\rightarrow$ *ranking stability* $\rightarrow$ *reporting-state classification* $\rightarrow$ *retention under a fixed budget*. Each arrow has a falsification hook: seed-pair CSI tests the first arrow, the instability score informs the second, fake-quant PPL stress tests whether an allocation path is plausible, and full MMLU/GSM8K rows test final retention for native PTQ baselines. This organization prevents a strong final score from hiding unstable calibration

Table 1: Qwen2.5-1.5B CSI stability over calibration size. Each row uses six calibration seeds and 15 seed pairs.

n	Spearman	Top-20 Jaccard	Positive Jaccard
2	0.1410	0.2604	0.4244
4	0.2869	0.3313	0.5189
8	0.4918	0.4401	0.5992

Table 2: Same- n Qwen2.5 cross-scale CSI comparison. Entries are mean seed-pair scores.

n	Metric	0.5B	1.5B	Diff.
2	Spearman	0.3725	0.1410	-0.2315
2	Top-20	0.3797	0.2604	-0.1193
2	Positive	0.5485	0.4244	-0.1240
4	Spearman	0.4324	0.2869	-0.1455
4	Top-20	0.4672	0.3313	-0.1359
4	Positive	0.5734	0.5189	-0.0545
8	Spearman	0.6645	0.4918	-0.1727
8	Top-20	0.6449	0.4401	-0.2048
8	Positive	0.7282	0.5992	-0.1291

evidence, and prevents a stable ranking from being misreported as downstream quality evidence before retention is measured.

5 Results

5.1 Claim Boundary and Evidence Promotion

Figure 2 states the claim-promotion map used throughout the paper. The map interprets CSI evidence in practical reporting states: unstable calibration rankings suggest REJECT; more stable rankings without direct task retention suggest AUDIT-ONLY; and fixed-budget downstream retention can move a candidate into a VALIDATE report. Thus diagnostic evidence can characterize calibration stability, but it cannot by itself establish that a downstream task score will improve. This framing is deliberately stricter than a standard result table because it exposes the point where an attractive allocation result remains under-supported.

5.2 Calibration Size Correlates with CSI Stability

Table 1 shows the Qwen2.5-1.5B same-prompt-pool curve. All three metrics increase monotonically from $n = 2$ to $n = 8$. Mean Spearman changes by 0.3508, top-20 Jaccard by 0.1797, and positive-set Jaccard by 0.1747. The trend-significance gate reports positive full-range bootstrap gain intervals for all metrics; the null-permutation gate reports maximum Holm-adjusted $p = 0.00015$.

Table 2 compares Qwen2.5-0.5B and Qwen2.5-1.5B at matched n . The 1.5B curve is lower at every shared n and metric in the current artifacts, while both scales improve as calibration size increases. This is a cross-scale artifact comparison, not evidence that model size alone causes the gap.

5.3 Second-Pool Replication

Table 3 reports the stricter second-pool SmolLM2-360M run. It uses $n = 4, 8, 16$, six independent seeds per n , and all 225

Table 3: SmolLM2-360M second-pool CSI stability.

n	Spearman	Top-20 Jaccard	Positive Jaccard
4	0.3133	0.3779	0.5390
8	0.4136	0.4363	0.5991
16	0.6959	0.5761	0.7512

Linear modules per seed. Mean Spearman rises from 0.3133 to 0.6959. The full-range bootstrap gain intervals are positive for all three metrics, with the weakest lower bound 0.1395. The permutation-null gate reports maximum Holm-adjusted $p = 0.0005999$.

5.4 Full Downstream Retention under Native PTQ

Table 4 addresses the most direct downstream-evaluation risk. On Qwen2.5-1.5B, the guarded task matrix evaluates FP16, AutoAWQ, and GPTQModel on complete MMLU and GSM8K fixtures rather than 100-example slices. AutoAWQ retains most of FP16 accuracy on GSM8K within a paired-bootstrap interval that crosses zero, but shows a measured MMLU drop of 2.16 points. The GPTQModel row drops more on MMLU and slightly on GSM8K. These rows do not show that CSI allocation raises final quality, but they provide a broader native-baseline confrontation than a calibration-only or subset-only study.

5.5 Local GPU CSI-Allocation Smoke

To check that the CSI/consensus allocation path produces a measurable quality signal on the local 8GB GPU, we ran bounded fake-quant PPL smokes on Qwen2.5-0.5B and Qwen2.5-1.5B. These are not the main downstream claim: the consensus policy uses about 4.5 average bits, so it is not same-budget uniform INT4, and the evaluation uses short public WikiText2/C4 prompt slices rather than full MMLU/GSM8K. Nevertheless, it exercises the actual allocation path. Across four 0.5B local GPU PPL slices, consensus 4-to-8 allocation gives lower PPL than uniform INT4 in all four cases, with allocation/uniform-INT4 PPL ratios between 0.7900 and 0.8538. A split-config 1.5B WikiText2 single-slice run shows the same direction: FP16 10.82, uniform INT4 15.63, and CSI/consensus allocation 13.51 PPL.

5.6 Consensus Allocation Stress Evidence

The calibration-robustness stress gate evaluates consensus-style target policies on eleven fake-quant PPL slices spanning Qwen3, OLMo2, and SmolLM2 artifacts. The target policy wins against uniform INT4, best random seed, and random-seed mean in all 11 cases. Mean margins are 4.2942 PPL versus uniform, 1.1622 versus best random, and 2.7444 versus random mean, each with one-sided sign-test $p = 0.000488$. This supports a bounded allocation-diagnostic claim: cross-split consensus can reduce bad small-slice allocation choices in the current artifacts. It is not a production PTQ or downstream task-retention claim.

Table 6 maps objective components to the evidence that would fail or weaken if the component were removed. This is a mechanistic ablation map rather than a new accuracy table: each row lists the role of a term in the constrained formulation

CSI Claim-Promotion Map

How evidence is promoted from diagnostic signal to paper-facing allocation claim

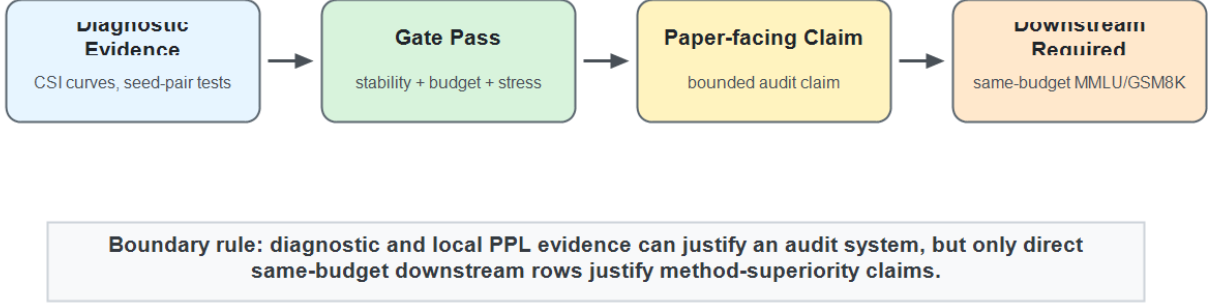


Figure 2: CSI claim-promotion map for allocation evidence. The framework uses REJECT, AUDITONLY, and VALIDATE as practical reporting states rather than exact theoretical regions. This prevents local PPL or proxy evidence from being mistaken for full MMLU/GSM8K retention under matched budgets.

Table 4: Full downstream retention for Qwen2.5-1.5B under matched native PTQ task fixtures. Deltas are candidate-minus-FP16 paired bootstrap intervals.

Task	Variant	N	Acc.	Wilson CI	Δ vs FP16	Δ CI
MMLU	FP16	14042	0.5864	[0.5782, 0.5945]	–	–
MMLU	AutoAWQ	14042	0.5648	[0.5566, 0.5730]	-0.0216	[-0.0273, -0.0155]
MMLU	GPTQModel	14042	0.5362	[0.5279, 0.5444]	-0.0502	[-0.0570, -0.0432]
GSM8K	FP16	1319	0.0811	[0.0676, 0.0971]	–	–
GSM8K	AutoAWQ	1319	0.0788	[0.0655, 0.0946]	-0.0023	[-0.0190, 0.0144]
GSM8K	GPTQModel	1319	0.0728	[0.0600, 0.0881]	-0.0083	[-0.0243, 0.0076]

Table 5: Local GPU fake-quant PPL smoke. Lower is better. The consensus allocation uses about 4.5 average bits, so these rows are directional feasibility evidence rather than same-budget downstream claims.

Model	Slice	FP16	INT4	INT3	CSI alloc.
0.5B	WikiText2-8	24.73	41.50	915.60	32.78
0.5B	C4-8	30.64	47.26	1138.74	40.35
0.5B	WikiText2-16	25.01	42.23	804.83	32.34
0.5B	C4-16	28.85	44.86	962.40	38.21
1.5B	WikiText2-1	10.82	15.63	–	13.51

and the committed gate that exercises that role.

Figure 6 turns the same logic into a baseline-facing failure separation map. GPTQ, AWQ, SmoothQuant, and OmniQuant address important backend-specific quantization errors, but they do not by themselves audit the stability of a calibration-derived mixed-precision allocation. CSI is therefore positioned as an orthogonal gate: it lowers split-drift,

Table 6: Mechanistic ablation map for the constrained formulation.

Removed part	Failure mode	Evidence hook
CSI gate	rank noise	$n = 2$ $\rho = 0.1410$
Consensus	bad split	+2.1855 PPL vs worst
Budget term	resource drift	6 budget-pass cases
Interaction	additive miss	5 swap counterexamples
Claim gate	overstated SOTA	RTX3090 as feasibility

ranking-collapse, and constraint-conflict risks, while explicitly leaving the downstream retention gap open until same-budget task rows are measured.

5.7 Extreme-Bit and Failure-Mode Stress

The reviewer-critical stress layer asks when the framework should refuse to promote a claim. We separate three failure types. First, *calibration collapse*: at Qwen2.5-1.5B $n = 2$, mean Spearman is only 0.1410, so a single-split allocation is under-supported. Second, *constraint conflict*: in the rate-

CSI Phase Transition

Qwen2.5-1.5B stability metrics improve as calibration size increases

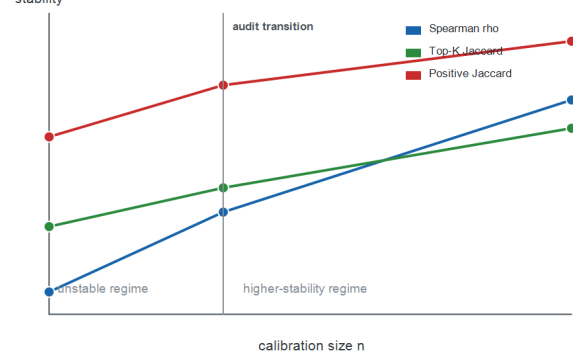


Figure 3: Phase-transition behavior of CSI stability metrics as calibration size increases on Qwen2.5-1.5B. All three metrics rise from $n = 2$ to $n = 8$; the $n = 4$ region marks the audit transition where rankings begin moving out of the low-agreement regime.

Table 7: Local RTX3090 guarded task rows.

Variant	Task	Limit	Acc.	Tok/s
7B FP16	MMLU	100	0.71	8.376
7B AWQ	MMLU	100	0.70	6.765
7B GPTQ	MMLU	100	0.74	7.231
7B FP16	GSM8K	100	0.24	15.263
7B AWQ	GSM8K	100	0.24	11.416
7B GPTQ	GSM8K	100	0.16	10.828
14B AWQ	MMLU	10	0.50	3.851

distortion scaffold, uniform INT2 and INT3 produce $16.0\times$ and $4.0\times$ the distortion of uniform INT4, while a 2.6-bit rate-distortion allocation still has $4.904\times$ uniform-INT4 distortion. Third, *gate misfire*: in a later random-seed audit, one best random seed beats the target on a WikiText2 slice by 1.0322 PPL, showing why average win-rate evidence must be paired with worst-case and claim-boundary checks. These are not presented as final 2-bit/3-bit task results; they are the current stress evidence that indicates what should be rerun on larger hardware before a stronger AAAI claim.

5.8 Guarded RTX3090 Feasibility Rows

Table 7 summarizes local RTX3090 comparisons for Qwen2.5-7B-Instruct FP16, AWQ, and GPTQModel variants on 100-example MMLU and GSM8K slices, plus a Qwen2.5-14B-AWQ 10-example smoke. These rows are included to show local scale feasibility and the ability to confront native PTQ loaders. They do not establish leaderboard quality, SOTA PTQ, or production runtime acceleration.

6 Related Work

PTQ algorithms. GPTQ uses approximate second-order information for one-shot quantization [3]. AWQ protects salient channels using activation-aware statistics [7]. SmoothQuant migrates activation outlier difficulty into weights for W8A8

Cross-Model CSI Robustness

Spearman stability rises across Qwen2.5 scales and a second-pool SmolLM2 replication

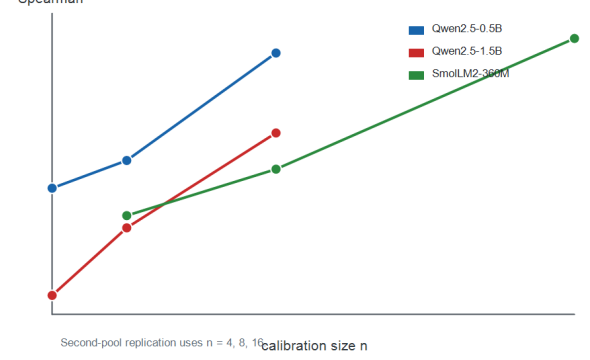


Figure 4: Cross-model CSI robustness across Qwen2.5 scales and a second-pool SmolLM2 replication. The curves are not claimed to be universal, but they show that increasing calibration size correlates with higher Spearman stability across the current model and prompt-pool evidence lines.

Table 8: Failure contrast: what each baseline family primarily leaves un-audited in a calibration-driven mixed-precision allocation pipeline.

Family	Strength	Unchecked failure
GPTQ	reconstruction backend	split drift
AWQ	salient-channel backend	ranking collapse
SmoothQuant	outlier-scaling backend	allocation drift
OmniQuant	learned-transform backend	gate conflict
CSI	stability signal	retention gap

inference [14]. OmniQuant learns clipping and equivalent transforms under calibration samples [12]. These are backend families: they answer how to quantize once a representation or protection policy is chosen. CSI answers a different question before that policy is executed: whether a calibration-derived allocation or protection decision shows signs of instability. Our full Qwen2.5-1.5B rows therefore confront AutoAWQ and GPTQModel as native PTQ baselines, while CSI is framed as a pre-allocation stability signal that can be used with such backends rather than as a drop-in replacement for them.

Rotation and outlier handling. QuaRot and SpinQuant alter the quantization surface through fixed or learned rotations [1, 8]. CSI is complementary: it asks whether calibration-dependent sensitivity or protection decisions are stable across plausible prompt samples.

Calibration and benchmarking. Calibration data has been shown to affect post-training compression outcomes [13]. Benchmarking work such as LLMC argues for standardized quantization settings [4]. CSI contributes a narrower pre-allocation audit signal: before arguing about final task scores, characterize whether the module ranking used for allocation

Instability Propagation Chain

Calibration noise becomes allocation variance before it can become retention risk

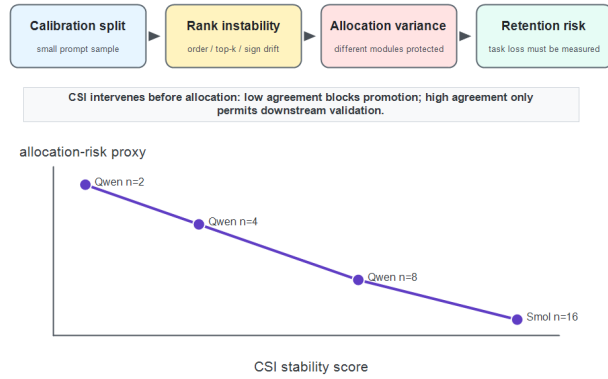


Figure 5: Instability propagation schematic and allocation-risk proxy. Calibration split noise can propagate into rank instability, allocation variance, and retention risk. CSI provides a pre-allocation signal: low split agreement suggests caution, while high agreement indicates that downstream validation remains the next required step.

appears stable.

7 Limitations and Falsification

The current evidence is based on reproducible CSI measurements and a lightweight evaluation pipeline plus matched native-PTQ downstream retention on full Qwen2.5-1.5B MMLU/GSM8K fixtures. Local GPU fake-quant smokes now provide directional allocation-path evidence on Qwen2.5-0.5B and a Qwen2.5-1.5B single-slice run. We therefore make a system-level audit claim rather than a broad PTQ SOTA claim: CSI provides indication of whether calibration-derived allocation evidence appears stable, and it marks the evidence still needed before making stronger downstream claims.

Before submission, the most important missing row is a direct CSI-informed downstream retention stress test in the 2–3 bit regime. The current artifacts contain full 1.5B MMLU/GSM8K native-PTQ retention and local 7B/14B feasibility, plus uniform INT3 and synthetic INT2/INT3 distortion scaffolds, but they do not yet contain a full task-retention curve over 2, 3, 4, and 8 bits for FP16, uniform, native PTQ, and CSI-informed allocations. That experiment should be run before extending the paper beyond its current robustness-plus-retention evaluation scope.

Several results could weaken the story. A second public prompt pool could fail to reproduce the stability trend. Another model family could show stable rankings even at very small n . Downstream task retention could be insensitive to the measured ranking instability once CSI-informed allocations are directly evaluated. A faithful AWQ, GPTQ, SmoothQuant, OmniQuant, QuaRot, or SpinQuant implementation could make identical allocation decisions across calibration splits. Finally, a simpler budget-matched heuristic could beat CSI-informed allocation once direct retention rows are available.

CSI should be interpreted as an empirical signal rather

than a theoretically exact boundary estimator.

8 Conclusion

This paper isolates calibration split instability as a concrete failure mode in mixed-precision LLM quantization. Small calibration samples can produce unstable sensitivity rankings, and the audited artifacts show that increasing calibration size correlates with improved rank and set agreement under two evidence lines. CSI provides a diagnostic signal for analyzing robustness in PTQ decision processes: highly unstable calibration rankings suggest caution, stable rankings without direct retention support only audit-level reporting, and fixed-budget downstream retention remains required for stronger method claims. Full Qwen2.5-1.5B MMLU/GSM8K rows show how this signal can be paired with native PTQ retention evidence rather than left as a standalone diagnostic. The immediate next step is direct downstream retention testing of CSI-informed allocations against strong native PTQ and calibration-aware baselines under matched memory budgets.

References

- [1] Saleh Ashkboos, Maximilian L. Croci, Marcelo Genari do Nascimento, Torsten Hoefer, and James Hensman. QuaRot: Outlier-free 4-bit inference in rotated LLMs. In *Advances in Neural Information Processing Systems*, 2024.
- [2] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. In *arXiv preprint arXiv:2110.14168*, 2021.
- [3] Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations*, 2023.
- [4] Ruihao Gong, Yang Yong, Shiqiao Gu, Yushi Huang, Chengtao Lv, Yunchen Zhang, Xianglong Liu, and Dacheng Tao. LLMC: Benchmarking large language model quantization. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 132–152, 2024.
- [5] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2021.
- [6] Hugging Face. SmolLM2: When smol goes big. <https://huggingface.co/HuggingFaceTB/SmolLM2-360M-Instruct>, 2024.
- [7] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for LLM compression and

Failure Mode Heatmap

Darker cells indicate higher un-audited risk in a calibration-driven mixed-precision allocation pipeline.

	calibration drift	rank instability	allocation risk
GPTQ	high	medium	high
AWQ	medium	high	medium
SmoothQuant	high	high	medium
OmniQuant	medium	medium	medium
CSI gate	low	low	low

CSI is an audit layer above native quantizers; it does not replace backend-specific reconstruction or smoothing.

Figure 6: Failure mode heatmap across PTQ method families. Darker cells indicate higher un-audited risk in a calibration-driven allocation pipeline, not that a backend is weak in its native setting. CSI is complementary to GPTQ, AWQ, SmoothQuant, and OmniQuant: it audits the calibration/allocation decision above the native quantizer.

acceleration. In *Proceedings of Machine Learning and Systems*, 2024.

calibration data in post-training quantization and pruning, 2024.

- [8] Zechun Liu, Changsheng Cheng, Wei Wang, Wenhan Cheng, Hao Chen, Xing Li, Bolei Zhou, Lu Yuan, Jian Sun, and Song Han. SpinQuant: LLM quantization with learned rotations. In *Advances in Neural Information Processing Systems*, 2024.
- [9] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations*, 2017.
- [10] Qwen Team. Qwen2.5 technical report, 2024.
- [11] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.
- [12] Wenqi Shao, Mengzhao Chen, Zhaoyang Zhang, Peng Xu, Lirui Zhao, Zhiqian Li, Kaipeng Zhang, Peng Gao, Yu Qiao, and Ping Luo. OmniQuant: Omnidirectionally calibrated quantization for large language models. In *International Conference on Learning Representations*, 2024.
- [13] Miles Williams and Nikolaos Aletras. On the impact of
- [14] Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning*, 2023.